

PREFACE

My aim is to show that o-minimal structures provide an excellent framework for developing tame topology, or *topologie modérée*, as outlined in Grothendieck's prophetic "Esquisse d'un Programme" of 1984. This close connection between tame topology and a subject created by model-theorists is hardly controversial, though perhaps not yet widely known or understood.

In the early 1980s I had noticed that many properties of semialgebraic sets and maps could be derived from a few simple axioms, essentially the axioms defining "o-minimal structures", as their models came to be called in an influential article by Pillay and Steinhorn. After Wilkie established in 1991 that the exponential field of real numbers is o-minimal the subject has grown rapidly. The supply of o-minimal structures on the real field is still increasing. In combination with general o-minimal finiteness theorems this gives rise to applications in real algebraic and real analytic geometry.

A rough version of this book was circulated informally in 1991, and was based on articles by various authors and on courses I gave at Stanford, Konstanz, and the University of Illinois at Urbana-Champaign. The main additions since then consist of Chapter 5 about the Vapnik-Chervonenkis property, Section 2 of Chapter 6 on fiberwise properties, Section 3 of Chapter 9 on a conjecture of Benedetti and Risler, and Chapter 10 on definable spaces. I have taken pains to present the subject in a way that is widely accessible and requires no knowledge of model theory. My initial ambition was to include more on the construction of o-minimal structures, but I decided to postpone this. (Traces remain as occasional references to what is supposed to happen "in the next volume".) Some material is included explicitly in the form of digressions or exercises requiring further background on the reader's part. Also, in solving a problem of Benedetti and Risler I quote without proof Wilkie's theorem of 1991. Otherwise everything is developed from scratch. Each chapter has notes at the end with references and comments.

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